

Sunny Zhang

ssunny.zhang@mail.utoronto.ca | [linkedin.com/in/sunnyzhang8](https://www.linkedin.com/in/sunnyzhang8) | sunnyzhang.dev

EDUCATION

University of Toronto, St. George

Honours BSc, Double Major in Computer Science and Mathematics

2025 – 2029

GPA: 3.7/4.0

PROJECTS

Feedback or Annealing? | *Python, NumPy, Numba, Matplotlib*

- Derived a geometric envelope for residual-driven aggregation from Bellman span contraction, creating an open-loop control that isolates feedback from the annealing schedule it induces.
- Evaluated matched residual, geometric, and fixed-width methods on a 9,261-state MDP across 3 update mixes, 20 paired seeds, and 5 compute budgets; feedback improved error by at most 0.3% and worsened it by up to 14.5%.
- Showed that annealing changed error by up to 23.5% relative to fixed resolution while all final feedback comparisons remained within a preregistered ± 0.02 equivalence region.

Parallel State Aggregation | *Python, NumPy, Numba, Matplotlib*

- Built a multithreaded benchmark of value iteration and adaptive state aggregation across 40,000–1,000,000-state MDPs, 1–10 threads, and paired seeds.
- Found value iteration reached the fixed ℓ_∞ error target 1.3–2.2 $\times$ faster across all 5 configurations; traced scaling limits to memory bandwidth and serial rebinning and exposed 99.67% uncounted work.

LaunchPilot | *Next.js, TypeScript, FastAPI, Postgres*

- Built an AI launch copilot in 72 hours with a 2-person team at Hack Canada 2026; won 2 tracks, the sole Backboard.io Best Use prize (1 of 59 teams) and placed top 3 of 94 in the SPUR startup track.
- Designed a supervised multi-agent workflow converting project briefs or GitHub repositories into market research, ICPs, positioning, messaging, KPIs, execution plans, and outreach drafts.
- Orchestrated three 5-agent stages with persistent threads and memory snapshots, gated by GitHub context, Auth0-scoped permissions, and human approval before execution.

EXPERIENCE

LLM/ML Research Intern

Jun 2026 – Present

Algoverse AI Research Program

- Leading research on hierarchical alignment in a multi-agent LLM trading environment, studying whether private reasoning predicts future misconduct.
- Designed a 4-source ablation study using regularized logistic regression to measure the predictive value of hidden reasoning within each individual hierarchy.
- Evaluating AUPRC, AUROC, calibration, and fixed-budget recall with run-level splits and held-out pressure settings.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL

Machine Learning and Data: PyTorch, Gymnasium, NumPy, Pandas

Framework & Libraries: FastAPI, Next.js, Numba, Flask, D3

Tools and Platforms: Git, Docker, Supabase, Hugging Face, Vercel